The chain

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Chapter 16 The contest

Who Controls the Chain

Since 2018 the United States has used export controls to limit China's access to advanced chips and chipmaking tools, and China has been learning to make both.

Specimen 16 · Who Controls the Chain

The world, with nineteen chokepoint towns pinned in garnet. Five towns building alternatives to those chokepoints are pinned in ink: Beijing, Wuhan, Shenzhen, Hefei and Dresden. The taller the garnet pin, the tighter the chokepoint at that town's stage of the chain. The pins are not to scale. If the globe were the size of the Earth, the tallest pin would stand about 1,000 km high.

In plain terms

Export controls work only when a country's own firms sell something the buyer cannot get elsewhere, so few governments can use them to much effect. An export control is a rule that requires the government's permission for a sale. The government names an item and a country or a buyer. From then on its own companies must hold a license before they ship that item to that country or buyer. The American government extends some of these rules to goods made abroad with American technology. The machines, software and chemicals used to make an advanced chip come from a short list of firms in the United States, the Netherlands and Japan, so those three governments can control who gets them.

In short

Japan and the Netherlands have aligned their own export controls with the American rules on chipmaking equipment[1]. China has answered by building its own tools and by restricting sales of US chips in China. Chinese toolmakers supplied 35 percent of China's own chipmaking equipment market in 2025, up from 10 to 15 percent before 2022[2]. Nvidia says the Chinese government restricted the

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

sales of its H200 chip that the United States licensed in February 2026[3].

The most advanced chips come mainly from Taiwan. It held 69 percent of the world's capacity for logic chips below 10 nanometers in 2022, by the count of the Semiconductor Industry Association and Boston Consulting Group[4].

01

The shape of the contest

China cannot yet make the most advanced lithography machines, which print a chip's finest circuits[5]. China's leading lithography maker, SMEE, produces only older, less precise machines, among them some deep-ultraviolet types[5]. China has also yet to show it can make high-bandwidth memory, the stacked memory beside an AI chip, in volume.

In 2022 the United States changed its aim from staying ahead of China to pushing China's chipmaking back. In September 2022 US National Security Advisor Jake Sullivan said the United States must keep "as large of a lead as possible" in foundational technologies such as advanced logic and memory chips[6]. The export rules of October 2022 were the first to apply his doctrine of a small yard with a high fence, meaning strict controls on a narrow set of technologies[1]. Those rules went beyond the old aim of staying two technology nodes ahead of China and tried to push China's capability back below the level it had already reached[7].

US license rules can target a single buyer or a whole country, and in 2022 the rules on advanced AI chips and the tools for advanced fabs began to cover all of China. In October 2018, for example, the Bureau of Industry and Security (BIS), the US Commerce Department's export control office, put the Chinese chip company Fujian Jinhua Integrated Circuit on its Entity List[8]. From then on any item under US export rules needed a license to reach the company[8]. The 2022 rules went further and required a license for any advanced AI chip, and any tool

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

3 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

5 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

6 Remarks by National Security Advisor Jake Sullivan at the Special Competitive... The White House, 16 September 2022. bidenwhitehouse.archives.gov

7 Choking off China's Access to the Future of AI. Center for Strategic and International Studies, 11 October 2022. csis.org

8 Addition of an Entity to the Entity List. Federal Register, 30 October 2018. federalregister.gov

for an advanced fab, bound for China[9]. BIS reviewed most applications under a presumption of denial, meaning it would normally refuse them[9]. It reviewed applications for equipment bound for the China fabs of firms headquartered in the United States or allied and partner countries case by case[9].

02

Chokepoints and who holds them

Segment	Leader	Share	Basis
EUV lithography	Netherlands	100.0%	revenue, 2025
Immersion DUV lithography	Netherlands	99.3%	revenue, 2025
Metrology and inspection	United States	71.7%	revenue, 2025
Deposition	United States	60.3%	revenue, 2025
Etch and clean	United States	52.6%	revenue, 2025
Photoresist	Japan	78%	revenue, 2023
Silicon wafers	Japan	53%	revenue, 2023
Leading-edge logic, below 10 nm	Taiwan	69%	capacity, 2022
Foundry, all nodes	Taiwan	81.9%	market share, 2025
HBM	South Korea	82.5%	revenue, 2025: SK hynix 63.2, Samsung 19.3, Micron 17.4
Logic chip design	United States	65%	sales, 2022
Mature logic, 28 nm and above	China	33%	capacity, 2022

No country leads every segment. The United States leads four of the table's segments. The Netherlands, Japan and Taiwan lead two each, and South Korea and China one each.

- **Equipment.** TechInsights 2025 revenue by parent-company headquarters, from CSET's ETO Supply Chain Explorer[10].
- **Materials.** Fuji Keizai's survey by parent-company headquarters, as Japan's Ministry of Economy, Trade and Industry published it in December 2025[11].

9 Implementation of Additional Export Controls: Certain Advanced Computing and… Federal Register, 13 October 2022. federalregister.gov

10 Documentation: Semiconductor Supply Chain Dataset. Emerging Technology Observatory (ETO), accessed 9 September 2026. eto.tech

11 METI, Semiconductor and digital industry strategy: future direction… Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

- **Memory.** Figures from the market research firm IDC, as SK hynix's US listing prospectus relays them[12].
- **Logic capacity.** The Semiconductor Industry Association and Boston Consulting Group's 2022 measurement of logic capacity, below 10 nm for Taiwan and at 28 nm and above for China[4].

Advanced packaging is missing from the table because no public source splits its capacity by country. The Taiwanese chipmaker TSMC does not publish one either.

03

The US export-control campaign, 2018 to today

US chip controls grew tighter from 2020 to 2024. In August 2020 BIS widened its Foreign Direct Product Rule to require a license for any foreign-made item produced with US technology and bound for Huawei[13]. In 2022 BIS blocked China's access to the most advanced AI chips and to the tools to make them[1]. It tightened those rules in 2023 and 2024[1]. But in August 2025 the US government began licensing some sales of Nvidia's H20 chip to China[14]. In February 2026 it began licensing small amounts of the H200[14].

The United States has also approved large AI chip sales in the Gulf. In November 2025 Commerce authorized G42 of the United Arab Emirates and HUMAIN of Saudi Arabia to buy the equivalent of up to 35,000 Nvidia GB300 chips each under security and reporting conditions[15]. On 10 July 2026 BIS moved the United Arab Emirates into Country Group A:5, the export-control group that holds close US allies[16]. It named the Emirati government, G42 and eight US-headquartered AI companies as recipients that need no license for controlled AI chips and systems[16].

The Export Controls page has the full timeline, with the legal citation for each rule.

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from… Center for Strategic and International Studies, 12 December 2024. csis.org

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

12 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

13 Addition of Huawei Non-U.S. Affiliates to the Entity List, the Removal of… Federal Register, 20 August 2020. federalregister.gov

14 NVIDIA CORP, Form 10-K annual report for the period ended 2026-01-25 (10-K). U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

15 Statement on UAE and Saudi Chip Exports. U.S. Department of Commerce, November 2025. commerce.gov

16 Enhanced Favorable Treatment for the United Arab Emirates Under the Export… Federal Register, 14 July 2026. federalregister.gov

Figure 16.1

US chip export controls tightened through January 2025, then partly eased

Eight US actions that shaped chip export controls

- 2019-05-16
  Huawei and 68 affiliates added to the Entity List
- 2022-10-07
  China-wide controls on AI chips and advanced-fab tools
- 2023-10-17
  Performance-density test added, covering Nvidia's A800 and H800
- 2024-12-02
  24 tool types, HBM controls, 140 entity listings
- 2025-01-13
  AI Diffusion rule announced, three country tiers
- 2025-05-13
  BIS stops enforcing the AI Diffusion rule
- 2025-08-11
  US officials expect 15% or more of licensed H20 revenue
- 2026-01-15
  H200 and MI325X move to case-by-case review

Source Federal Register rules, BIS press releases and Nvidia filings, cited for each entry on the Export Controls timeline

04

Ways around the controls

Firms can use controlled chips without any chip crossing a border, because renting out time on a chip does not count as exporting it[17]. A Singapore data center selling time on an H200 already installed there, for example, moves nothing across a border[17]. The AI Diffusion rule of January 2025 required a license for advanced AI chips bound anywhere in the world[18]. For the data centers it approved, it said that API access to a model, and rented capacity for inference, "are not prohibited"[18]. On 13 May 2025 the Commerce Department said it would rescind the rule and told BIS officials not to enforce it[19].

The American rules reach that trade through the firms holding the chips. A license under the 15 January 2026 rule for H200-class chips bound for China or Macau requires the applicant to obtain a

17 15 CFR 734.13 -- Export. Electronic Code of Federal Regulations, 3 June 2016. ecfr.gov

18 Framework for Artificial Intelligence Diffusion. Federal Register, 15 January 2025. federalregister.gov

19 Department of Commerce Announces Rescission of Biden-Era Artificial... U.S. Bureau of Industry and Security, 13 May 2025. bis.gov

description of the recipient's know-your-customer procedures[20]. These are the recipient's checks on who its customers are. The applicant must also obtain the recipient's list of customers renting computing power in China and seven other places, among them Russia and Iran[20]. The recipient must also promise that no algorithm trained on the chips is served to blacklisted or military end users[20].

Smugglers route chips past the controls through Southeast Asia. Since 14 July 2025 Malaysia has required a permit to export, transship or transit any high-performance US-origin AI chip[21]. On 25 March 2026 the Justice Department announced charges against three people accused of using Thai companies' names to order servers with export-controlled chips for China[22]. The orders included one for $170 million that was never completed[22].

In July 2025 the US House of Representatives' select committee on China cited estimates that a median of 140,000 chips were smuggled to China in 2024[23]. A bill in Congress, the Chip Security Act, would require location verification on controlled chips within 180 days of enactment[24]. The House Foreign Affairs Committee approved the House version 42-0 on 26 March 2026[25].

05

Allies: the Netherlands, Japan, Korea and Europe

US tool controls alone would only shift the sales to Dutch and Japanese suppliers, so the campaign depends on those two governments. Both have aligned their own controls on chipmaking equipment with the American rules[1].

- **The Netherlands.** The Dutch government began requiring export licenses for advanced deep-ultraviolet tools on 1 September 2023 and widened the controls in September 2024[26]. On 1 April 2025 it extended them to measurement and inspection tools[27].

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from... Center for Strategic and International Studies, 12 December 2024. csis.org

20 Revision to License Review Policy for Advanced Computing Commodities. U.S. Government Publishing Office, 15 January 2026. govinfo.gov

21 Export, transshipment and transit of high-performance AI chips of US origin now... Ministry of Investment, Trade and Industry, 14 July 2025. miti.gov.my

22 Chinese National and Two U.S. Citizens Charged with Conspiring to Smuggle... U.S. Department of Justice, 25 March 2026. justice.gov

23 Protecting U.S. Tech: China Committee and Bipartisan, Bicameral Leaders Unite... U.S. House Select Committee on the Chinese Communist Party, 7 July 2025. chinaselectcommittee.house.gov

24 S.1705 – Chip Security Act, 119th Congress (2025-2026), bill text. U.S. Congress, 8 May 2025. congress.gov

25 UNANIMOUS: Bipartisan Huizenga Legislation to Curb AI Chip Smuggling Passes... U.S. House of Representatives, 26 March 2026. huizenga.house.gov

26 The Netherlands expands export control measure for advanced semiconductor... Government of the Netherlands, 6 September 2024. government.nl

27 Klever: export controls on advanced semiconductor manufacturing equipment to be... Government of the Netherlands, 15 January 2025. government.nl

- **Japan.** Japan's controls on advanced manufacturing equipment took effect on 23 July 2023, apply to all destinations and name no country[28].
- **US rules for South Korean fabs in China.** BIS revoked the Validated End-User status that let Samsung's and SK hynix's China subsidiaries and Intel Semiconductor (Dalian) receive US equipment without individual licenses, effective 31 December 2025[29]. BIS said it would license those fabs to keep running, but not to expand capacity or upgrade technology[30]. SK hynix, which bought the Dalian fab from Intel, received an annual BIS license in December 2025 to ship approved US equipment to its China fabs during 2026[31].

A BIS rule effective 29 September 2025 extended Entity List restrictions to any company at least 50 percent owned by one or more listed firms[32]. The rule reached Nexperia, a Nijmegen chipmaker that supplies Europe's car industry, and the Dutch government said it worked to secure an exemption for the company[33]. On 30 September the Dutch economy minister invoked the Goods Availability Act against Nexperia[34]. He cited governance failures that risked moving technology out of Europe[34]. The order aimed to keep Nexperia's production capabilities available to Europe in an emergency[33].

On 4 October the Chinese authorities imposed an export control on Nexperia's operations in China, so deliveries from those operations to customers needed an export license[35]. The Dutch government said uncertainty over licensing and long processing times caused chip shortages among Nexperia's customers and stopped production at some companies[35].

Both the US rule and the Dutch order were suspended in November. On 10 November BIS suspended its rule until 9 November 2026[36]. The minister suspended his order on 19 November[33]. He cited constructive meetings with the Chinese authorities and the measures they had taken to ensure the supply of chips to Europe and the rest of

28 CSIS Translation: Updated Japanese Export Controls on High-Performance... Center for Strategic and International Studies, 18 July 2023. csis.org

29 Revocation of Validated End-User Authorizations in the People's Republic of... Federal Register, 2 September 2025. federalregister.gov

30 Department of Commerce Closes Export Controls Loophole for Foreign-Owned... U.S. Bureau of Industry and Security, 29 August 2025. bis.gov

31 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

32 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register, 30 September 2025. federalregister.gov

33 Update on invoking Goods Availability Act - 19 november 2025. Government of the Netherlands, 19 November 2025. government.nl

34 Minister of Economic Affairs invokes Goods Availability Act. Government of the Netherlands, 12 October 2025. government.nl

35 government.nl

36 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain... Federal Register, 12 November 2025. federalregister.gov

the world[33]. Europe holds a chokepoint in leading-edge lithography, but none in the mature-node chips its car industry uses.

A bill in the US Congress would push allies to match US limits on chipmaking tools sold to China. The MATCH Act, a bill introduced on 8 April 2026, would ban sales of immersion deep-ultraviolet scanners to China[37]. It would also restrict exports to, and servicing of, every chipmaking plant run by five Chinese companies, CXMT, Hua Hong, Huawei, SMIC and YMTC[37]. It would give allies 150 days to match those limits before the United States extends its own controls to foreign-made items that use US software, technology or components[37].

06

China's response

China has answered by replacing foreign chips at home, restricting some American suppliers, limiting its exports of minerals and building its own chipmaking tools. In 2024 China's Ministry of Industry and Information Technology reportedly told Chinese telecom operators to strip foreign chips out of their networks by 2027[2].

China has also taken measures against American suppliers. Micron, the American memory maker, failed a review by the Cyberspace Administration of China (CAC) on 21 May 2023 and lost its critical-infrastructure buyers in China[38]. In September 2025 the CAC reportedly told firms including ByteDance and Alibaba to cancel Nvidia orders[39]. Nvidia says the Chinese government restricted the H200 sales that the United States licensed in February 2026[3]. It says China had earlier restricted licensed H20 sales too[3].

China's Ministry of Commerce (MOFCOM) has also restricted exports of minerals.

- **3 December 2024.** MOFCOM banned exports of gallium, germanium, antimony and superhard materials to the United States, one day after the American chip rules[40].

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

3 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission, 26 August 2026. sec.gov

33 Update on invoking Goods Availability Act - 19 november 2025. Government of the Netherlands, 19 November 2025. government.nl

37 Risch, Ricketts, Kim Introduce MATCH Act; Level the Global Playing Field for... U.S. Senate Committee on Foreign Relations, 8 April 2026. foreign.senate.gov

38 Micron Aggression: The Right Response to Beijing's Ban on the U.S. Chipmaker. Center for Strategic and International Studies, 22 June 2023. csis.org

39 Beijing blocks Nvidia sales in vote of confidence for Chinese chipmakers, the... Center for Security and Emerging Technology, 18 September 2025. cset.georgetown.edu

40 Ministry of Commerce Notice 2024 No. 46: Notice Concerning Strengthening... Center for Security and Emerging Technology, 3 December 2024. cset.georgetown.edu

- **4 April 2025.** It required export licenses for seven medium and heavy rare earths[41].
- **9 October 2025.** It announced that from 1 December 2025 foreign firms would need its license to export listed goods made abroad with Chinese rare earths worth 0.1 percent or more of their value[42]. MOFCOM suspended that rule on 7 November 2025 until 10 November 2026[43].
- **9 November 2025.** MOFCOM suspended its ban on exports of gallium, germanium, antimony and superhard materials to the United States until 27 November 2026[44]. It acted after Donald Trump and Xi Jinping met at the Asia-Pacific summit in October 2025[45].

China's suspension of its October 2025 controls did not cover the April 2025 licensing rules for seven rare earths[43].

Chinese toolmakers are winning a growing share of their home market. They held 10 to 15 percent of it before 2022, 25 percent in 2024 and 35 percent in 2025[2]. They hold about 40 percent of the Chinese market for etch and deposition tools[2]. But in September 2025 the Center for Strategic and International Studies (CSIS) judged that making its own extreme-ultraviolet machines remained firmly out of China's reach despite massive state investment[5].

Chinese chipmakers remain behind at the leading edge, where TSMC makes 2 nm chips[46]. SMIC's most advanced process, N+3, reaches the transistor density of TSMC's 6 nm-class process without extreme-ultraviolet scanners[47]. It prints its tightest layers in several passes on deep-ultraviolet tools[47].

In September 2025 the research firm SemiAnalysis conservatively put SMIC's capacity at 7 nm and below at 45,000 wafers a month by the end of 2025[48]. It expected 60,000 in 2026 and 80,000 in 2027[48]. Those figures count wafers, and how many working chips they give depends on yield, the share of chips on a wafer that work.

2 China’s Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

5 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements. Center for Strategic and International Studies, 24 September 2025. csis.org

41 Announcement No.18 of 2025 of The Ministry of Commerce and The General... Ministry of Commerce of the People's Republic of China, 4 April 2025. english.mofcom.gov.cn

42 Ministry of Commerce Notice 2025 No. 61: Announcement of the Decision to... Center for Security and Emerging Technology, 9 October 2025. cset.georgetown.edu

43 MOFCOM and GAC Announcement 2025 No. 70, suspending the October 2025... Ministry of Commerce of the People's Republic of China, 7 November 2025. mofcom.gov.cn

44 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

45 The U.S.-China Trade Truce Has Not Solved the Gallium Problem. Center for Strategic and International Studies, 11 May 2026. csis.org

46 Why Taiwan Fears ‘America First’ Risks Eroding Its ‘Silicon Shield’. Stimson Center, 10 October 2025. stimson.org

47 Is SMIC N+3’s Metal Pitch Smaller than Intel 18A’s? SemiAnalysis, 14 June 2026. newsletter.semianalysis.com

48 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

SemiAnalysis judged that memory would limit China's output of AI chips before logic wafers did[48]. It expected Huawei to ship about 805,000 of its Ascend AI chips in 2025, with dies made by both TSMC and SMIC[48]. SemiAnalysis says Huawei got Ascend dies made at TSMC, in violation of export controls, while SMIC ramped up its own production[48]. By its count, China had procured roughly 13 million memory stacks, enough for about 1.6 million Ascend 910C packages[48]. It expected CXMT, China's own memory maker, to make only about 2 million stacks in 2026, enough for 250,000 to 300,000 more packages[48].

Huawei said in September 2025 that its Ascend 950 chips, due in 2026, would carry high-bandwidth memory of its own design[49]. It gave no figure for how much of that memory it would make[49].

Figure 16.2

China's revenue share fell at all four largest toolmakers after fiscal 2024

China's share of revenue at the four largest chip-tool makers, peak fiscal year and latest



Source ASML 2025 Form 20-F (total net sales), Applied Materials and Lam Research 10-K filings and Tokyo Electron FY2024 and FY2026 consolidated financial statements, each China share computed from the filing's sales by region [50]

07

Industrial policy: CHIPS, Rapidus, Big Fund and the rest

The US government helps pay for new chip plants with grants and a tax credit. The CHIPS and Science Act authorized about $52.7 billion for manufacturing incentives, research and workforce support[2]. By July 2025 Commerce had awarded $30.9 billion of direct funding to 19 companies for 40 projects[51]. The grants covered a median 14.2 percent

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

48 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

49 Groundbreaking SuperPoD Interconnect: Leading a New Paradigm for AI... Huawei, 18 September 2025. huawei.com

50 ASML HOLDING NV, Form 20-F annual report for the period ended 2025-12-31... U.S. Securities and Exchange Commission, 25 February 2026. sec.gov

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

of each project's capital cost[51]. Commerce took a federal tax credit, the Advanced Manufacturing Investment Credit, into account when it set the size of those grants[51]. Since the One Big Beautiful Bill Act of 2025, the credit covers 35 percent of qualified investment in an advanced chip plant placed in service after 2025, up from 25 percent[51].

On 22 August 2025 the US government agreed to take a 9.9 percent stake in Intel by buying $8.9 billion of its stock[52]. Of that sum, $5.7 billion was to come from CHIPS Act grants that Intel had been awarded but not yet paid[52]. The other $3.2 billion was to come from Intel's award under Secure Enclave, a program to supply trusted and secure chips to the US Department of Defense[52].

The United States has also used tariffs to draw chip investment. In January 2026 the American Institute in Taiwan and the Taipei Economic and Cultural Representative Office in the United States signed a memorandum on investment[53]. Under it, Taiwanese chip and technology firms are to invest $250 billion in US semiconductor, energy and AI production[53]. In the same memorandum the United States agreed to cap some of its national-security tariffs on Taiwanese goods at 15 percent[53]. On 16 July 2026 TSMC announced another $100 billion for four more Arizona fabs, bringing its total Arizona investment to $265 billion[54].

The European Commission says the EU still depends on other countries for advanced chip manufacturing[55]. On 3 June 2026 the Commission proposed a Chips Act 2.0 to support advanced chip production in the EU[55].

Japan backs a homegrown chipmaker, Rapidus. Rapidus began prototyping 2 nm gate-all-around transistors at its IIM-1 plant in July 2025[56]. In February 2026 it raised another 267.6 billion yen, about $1.7 billion, from Japan's Information-Technology Promotion Agency (IPA) and 32 companies, aiming for 2 nm mass production by 2027[57]. By June 2026 it had raised 424.95 billion yen in equity in all, after a further 150 billion yen from the IPA[58].

CSIS puts China's cumulative state funding for chips since 2014 at an estimated $150 billion and calls the CHIPS Act substantially smaller in

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen… U.S. Government Accountability Office, 11 December 2025. files.gao.gov

52 Intel and Trump Administration Reach Historic Agreement to Accelerate American… Intel, 22 August 2025. intc.com

53 Implementing Certain Tariff-Related Elements of a Trade and Security Agreement… Federal Register, 28 May 2026. federalregister.gov

54 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix, 16 July 2026. phoenix.gov

55 Proposal for the Chips Act 2.0. European Commission, 3 June 2026. digital-strategy.ec.europa.eu

56 Rapidus Achieves Significant Milestone at its State-of-the-Art Foundry with… Rapidus, 18 July 2025. rapidus.inc

57 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private… Rapidus, 27 February 2026. rapidus.inc

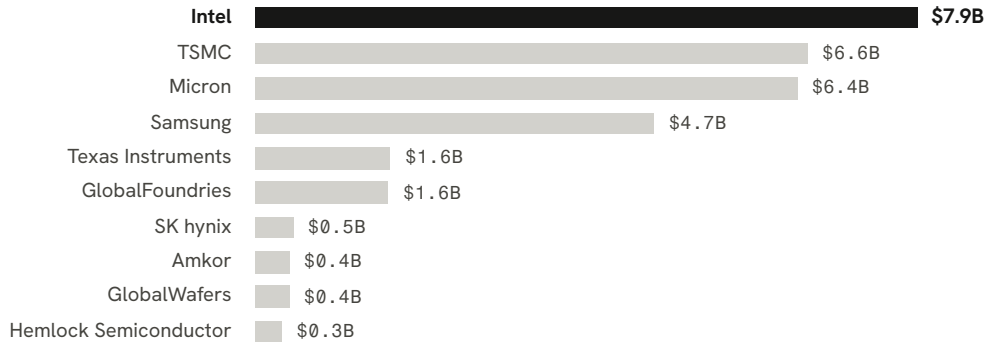
58 Rapidus Completes 150 Billion Yen Funding Round from Japan Government. Rapidus, 5 June 2026. rapidus.inc

scale[2]. The latest round of China's National Integrated Circuit Industry Investment Fund, the Big Fund, reached $47 billion[59]. That round, the fund's third, was set up in May 2024[60].

Figure 16.3

Intel led CHIPS Act direct funding with $7.865 billion by July 2025

CHIPS Act direct funding awarded, by firm, as of July 2025



Source GAO-26-107882, Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain [51]

08

Taiwan

The world relies on Taiwan for most of its advanced chips. Taiwan's foundries held 81.9 percent of the world's contract chipmaking market in 2025, according to the Taiwan Semiconductor Industry Association's August 2026 report[61]. TSMC makes its 2 nm chips only in Taiwan[46]. Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022[4].

TSMC's advanced packaging capacity could not keep up with demand. Asked in October 2025 how much CoWoS packaging capacity he would add, TSMC's chairman said only that the company was "working very hard to narrow the gap between the demand and supply"[62]. He called wafer and packaging capacity both "very tight"[62].

The silicon shield argument holds that this concentration deters an attack on Taiwan, because so many countries rely on its chips[46]. The

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

4 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group and Semiconductor Industry..., May 2024, p. 14. semiconductors.org

46 Why Taiwan Fears 'America First' Risks Eroding Its 'Silicon Shield'. Stimson Center, 10 October 2025. stimson.org

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

59 China's Evolving Industrial Policy for AI. RAND Corporation, 26 June 2025. rand.org

60 Six banks to invest in big way in IC fund. The State Council of the People's Republic of China, 29 May 2024. english.www.gov.cn

61 Overview on Taiwan Semiconductor Industry (2026 Edition). Taiwan Semiconductor Industry Association, 7 August 2026, p. 9. tsia.org.tw

62 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese... Refinitiv StreetEvents, 16 October 2025, p. 7. investor.tsmc.com

Stimson Center calls that open to debate, though many Taiwanese believe it[46].

With an eye to that shield, Taiwan's rules keep TSMC's newest process at home first[46]. As a result, TSMC's Arizona fabs have run a generation behind its fabs in Taiwan[46]. The Stimson Center expected 2 nm production in Arizona in 2028 at the earliest[46]. TSMC said in October 2025 that it was "preparing to upgrade our technologies faster to N2 and more advanced process technologies in Arizona"[63]. The four Arizona fabs it announced in July 2026 are to run at 2 nm or more advanced[54].

TSMC has also announced two advanced packaging fabs for Arizona[64]. But TSMC's work in Taiwan rests on a local supply chain of more than 300 firms, which the Stimson Center judged extremely difficult to copy in the United States[46]. The US push for chip plants at home has raised fears in Taiwan that the shield could weaken[46].

09

Who gains and who pays

- **Taiwan and Korea.** Fabs built abroad add capacity without taking any away from Taiwan and Korea. TSMC makes its 2 nm chips only in Taiwan[46]. Korean firms took 82.5 percent of the world's high-bandwidth memory revenue in 2025, by the market research firm IDC's count[12].
- **The Netherlands and Japan.** Their toolmakers build the controlled machines, so the two governments gain a say in who gets them. China accounts for a falling share of the firms' sales. At ASML, China's share of net system sales fell from 41 percent in 2024 to 33 percent in 2025[65]. That measure leaves out ASML's service and field option sales, which come from servicing and upgrading machines already installed[66]. Against total net sales, the measure in the chart, it fell from 36.1 percent to 29.1 percent[50]. At Tokyo Electron, China's share of sales fell from 44.4 percent in the year to March 2024 to 34.1 percent in the year to March 2026[67]. The company said

12 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission, 10 July 2026. sec.gov

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China's share of its sales dropped to 26.8 percent in the quarter to March 2026 because spending on leading-edge nodes grew faster than spending on mature nodes[68].

- **The United States.** The Semiconductor Industry Association, using data from World Semiconductor Trade Statistics, Gartner and Omdia, put US firms' share of all chip sales in 2025 at 53.4 percent, the highest since 1984[69]. US firms also took 65 percent of the world's logic chip sales in 2022[70]. Commerce expects the CHIPS-funded projects to lift the US share of global leading-edge logic chip manufacturing from 0 percent in 2022 to 20 percent by 2030[51].
- **China.** Chinese suppliers are replacing foreign ones at home, most visibly in mature-node and commodity chips[2]. In September 2025 SemiAnalysis expected China to make fewer Ascend chips in 2026 than in 2025 unless memory was smuggled in, because its own memory maker could supply too little[48].

Five questions to answer from memory

1. What does a country need for its export controls to work?
   see Chokepoints and who holds them
2. How do US export controls reach goods made in other countries?
   see The US export-control campaign, 2018 to today
3. What share of China's own chipmaking equipment market did Chinese toolmakers supply in 2025? see China's response
4. What share of ASML's system sales went to China in 2025?
   see Who gains and who pays
5. Where are most leading-edge chips made? see Taiwan

Five things to remember

- An export control works only when a country's firms sell something the buyer cannot get elsewhere.
- US controls also cover some foreign goods made with American technology.
- Chinese toolmakers' share of China's own chipmaking equipment market rose from 10 to 15 percent before 2022 to 35 percent in 2025.
- China's share of ASML's and Tokyo Electron's sales has fallen since 2024.

2 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip... Center for Strategic and International Studies, 24 March 2026. csis.org

48 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The... SemiAnalysis, 8 September 2025. newsletter.semianalysis.com

51 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen... U.S. Government Accountability Office, 11 December 2025. files.gao.gov

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70 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

— Taiwan held 69 percent of the world's logic capacity below 10 nm in 2022, and TSMC makes its 2 nm chips only there.

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